

ASSIGNMENT 1

Instructions for Students

- Create **separate .js files** for each question.
 - Use **comments** to explain your logic.
 - Test all programs using the **browser console** or **Node.js**.
 - Use **descriptive variable names**, indentation, and formatting.
 - Use only **core JavaScript** (no frameworks or libraries).
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Q1. Personalized Login Greeting

Create a program that greets users with time-based messages.

Requirements:

- Declare variables for `userName` and current hour using `new Date().getHours()`.
 - If time is before 12 → “Good Morning [name]!”
 - If time is between 12 and 17 → “Good Afternoon [name]!”
 - Otherwise → “Good Evening [name]!”
 - Use template literals and proper string formatting.
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Q2. Multi-Type Data Summary

Simulate a database summary of various user data entries.

Requirements:

- Declare variables of different data types: string, number, boolean, array, object, null, undefined.
 - Use `typeof` operator and `Array.isArray()` to identify each type.
 - Print a formatted report (label + value + type) using one `console.table()` statement.
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Q3. Monthly Expense Tracker

Develop a script that calculates total and average monthly expenses with dynamic updates.

Requirements:

- Use an array of expenses for 5 categories (food, travel, rent, bills, leisure).
 - Use arithmetic and assignment operators to add 10% tax to the total.
 - Use `toFixed(2)` to round values.
 - Display total, average, and final amount after tax.
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Q4. Academic Performance Evaluator

Check whether a student is promoted or not based on multiple subjects.

Requirements:

- Input marks of 5 subjects (use an array).
- Calculate average and overall percentage.
- Use logical operators to print:
 - “Promoted with Distinction” if $\geq 85\%$
 - “Promoted” if between 50%–84%
 - “Detained” if $< 50\%$

- Add validation: if any subject < 35 → automatically “Detained”.
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Q5. Weather Activity Planner

Advise a user on what to do based on temperature, rain, and wind speed.

Requirements:

- Create variables for temperature, isRaining, windSpeed.
 - If raining → “Stay indoors with hot coffee.”
 - Else if temperature > 35 → “Go swimming.”
 - Else if temperature < 15 and windSpeed > 20 → “Too cold and windy — stay home.”
 - Else → “Perfect day for a walk.”
 - Combine && and | | conditions smartly.
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Q6. Progressive Discount System

Build a system that applies dynamic discount tiers for shopping totals.

Requirements:

- Input total purchase amount.
- If total ≥ 10000 → 25% discount.
- Else if ≥ 5000 → 15% discount.
- Else if ≥ 2000 → 5% discount.
- Otherwise → no discount.
- Calculate and print:

- Original total
 - Discount percentage
 - Final price after discount
 - Use `Math.round()` for rounding.
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Q7. Smart Guessing Game (Number Range)

Enhance the guessing game by allowing range checking.

Requirements:

- Generate a random secret number between 1–50 using `Math.floor(Math.random() * 50) + 1`.
 - Let `userGuess` be a variable with a test value.
 - Print:
 - “Correct guess!” if equal
 - “Very close!” if within ± 3 of secret
 - “Too high” or “Too low” otherwise
 - Use nested `if` conditions and logical operators.
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Q8. Employee Salary Projection

Calculate a 5-year salary projection with yearly increments.

Requirements:

- Declare current salary and annual increment rate (in %).

- Use a for loop to compute salary for each of 5 years.
 - Print yearly salary in a formatted console table.
 - Apply rounding and assignment operators.
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Q9. Random Math Quiz Generator

Generate random arithmetic challenges and evaluate user answers automatically.

Requirements:

- Use `Math.random()` to generate two numbers (1–20) and a random operator from `['+', '-', '*', '/']`.
 - Calculate the correct answer using switch statement.
 - Display question and correct answer in the console.
 - Ensure division is rounded to two decimals with `toFixed(2)`.
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Q10. Citizen Eligibility Validator

Evaluate if a person can vote, drive, and apply for a passport based on age and citizenship.

Requirements:

- Input: age and `isCitizen` (boolean).
- Use nested if-else and logical operators to print:
 - “Eligible for all services.” if citizen and age ≥ 18
 - “Eligible to vote only.” if citizen and 18–20
 - “Only age criteria met.” if not citizen but age ≥ 18

- “Not eligible yet.” otherwise
- Use `console.log()` for all outcomes.